

SallyIP

The Verification-First Operating System for Intellectual Property Law

"General AI drafts prose. SallyIP proves law."

Replacing high-risk, hallucination-prone models with a deterministic, citation-verified intelligence engine for the \$65B+ global IP economy.

100%	0%	98.8%	400%
Citation Integrity Zero Fabricated Sources	Adversarial Error Deterministic Fail-Closed	Prior Art Recall @ k 100M+ Patent Databases	Margin Expansion Fixed-Fee Drafting Profit

Company: SallyIP Inc.	Target Raise: \$3.5M - \$5.0M	Stage: Seed / Series A
Sector: Vertical LegalTech / IP AI	Gross Margins: 78% - 84%	Deployment: Cloud / VPC / Word Add-in

Executive Thesis: Intellectual property law is the ultimate crucible for artificial intelligence. Patents and IP contracts protect multi-billion-dollar enterprise assets, but drafting and prosecution are plagued by a \$20B production bottleneck. Attorneys face intense fixed-fee compression (\$8,000-\$12,000 per patent) yet cannot deploy generic AI tools due to catastrophic hallucination rates (58%-82%). SallyIP introduces the industry's first verification-first architecture: combining hybrid multi-path retrieval, code-level citation integrity, and exact-quote verification to deliver attorney-grade patent drafting, Office Action response, and freedom-to-operate clearance.

Defensible Traction: 70/70 security test suites passing, 28/28 deterministic verification gates verified, and pilot-ready architecture. SallyIP converts 20-hour patent preparation marathons into rigorous, attorney-verified 3-hour workflows.

1. The Problem: The \$20B Patent Bottleneck & AI Malpractice Threat

Patent preparation and prosecution is an indispensable \$20B legal service, but its operating model is breaking down:

- **Extreme Production Costs:** Drafting a defensible patent requires 20-40 hours of partner and associate time, costing clients \$10,000 to \$25,000+ per application.
- **The Fixed-Fee Margin Squeeze:** Over 72% of corporate clients now mandate capped fixed fees (\$8,000-\$12,000), compressing law firm partner realization rates to historic lows.
- **25+ Month Examination Backlogs:** Over 700,000 applications are filed yearly with the USPTO alone. Attorneys spend over 60% of their billable hours on repetitive administrative chores: claim trees, antecedent basis checks, figure numeral tracking, and rejection dissection.
- **The AI Malpractice Threshold:** While firms urgently seek automation, general LLMs (GPT-4, Claude) hallucinate on 58%-82% of legal queries. Even first-gen legal RAG tools (Harvey, CoCounsel) suffer from 17%-33% error rates. In patent law, a single hallucinated prior art reference or defective claim limitation can trigger patent invalidation or ethical sanction.

GENERAL LLMs (GPT-4 / Claude) 58% - 82% Hallucination Fabricates court cases, statutes, and non-existent patent numbers.	FIRST-GEN LEGAL RAG (Harvey) 17% - 33% Hallucination Dangling citations, unverified quotes, and shallow broad summaries.	SALLYIP VERIFICATION-FIRST 0.0% Tolerated Hallucination Fail-closed architecture. 100% citation integrity & verbatim proof.
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2. The Solution: SallyIP's Verification-First Architecture

SallyIP does not rely on fragile prompt engineering. Instead, it places a deterministic, algorithmic verification barrier around model reasoning:

- **Multi-Path Hybrid Retrieval:** Fuses BM25 keyword matching, dense vector embeddings, and curated static jurisdiction packs (US, EPO, UK, India) via Reciprocal Rank Fusion (RRF). Zero results is treated as a valid state rather than forcing hallucinations.
- **Citation Integrity Guard:** Scans generated drafts for source tokens ([S1], [S2]). Any citation not grounded in retrieved text is expunged and flagged unverified.
- **Exact Verbatim Quote Verification:** Every generated quotation is matched against source documents. If altered by even a single word, quotation marks are stripped automatically.
- **Deterministic Fail-Closed Policy:** When primary authority is missing for high-risk conclusions (novelty, validity, infringement), Sally outputs: *'I could not verify this proposition from the available authorities.'*
- **Substantive Patent Standards Check:** Validates assertions against statutory rules under 35 U.S.C. 101 (Alice/Mayo two-step), 102 (single-reference anticipation), 103 (PHOSITA standard), and 112 (written description & enablement).

3. Full-Lifecycle Intellectual Property Product Modules

Module	Capabilities & Technology	Practitioner Impact
Novelty & Prior Art Radar	Limitation-by-limitation claim comparison across 100M+ USPTO, EPO, WIPO patents and NPL (arXiv, PubMed, IEEE).	98.8% Recall @ k Sub-2.4s search latency
Patent Drafting Studio	Synthesizes provisional (Section 111(b)) and nonprovisional (Section 111(a)) applications. Real-time antecedent basis checks and Section 101/112 validation.	6x Faster Drafting 100% Antecedent accuracy
Office Action Dossier	Deconstructs USPTO/EPO rejections. Evaluates examiner allowance rates, generates claim traversal arguments and interview talking points.	4.2x Faster Response Direct amendment simulator
FTO Clearance Radar	Maps product features directly to third-party claims. Automated technical design-around suggestions and claim maintenance fee tracking.	Zero Overlooked Claims Real-time landscape alerts
Trademark Clearance	Cross-register search (USPTO, EUIPO, Madrid). DuPont confusion factor analysis across Nice Classes 1-45.	94.2% Correlation Sub-1.8s clearance run
Claim Charts & EoU	Generates courtroom-ready Evidence of Use charts mapping patent claims to accused products, teardowns, and whitepapers.	Court-Ready Charts Prosecution estoppel audit
Invention Interviewer	Interactive 7-slot extraction engine capturing disclosures from inventors without requiring legal jargon. Adapts to user sophistication.	Max 4 Turns Plain language to patent claim

4. Proprietary Defensible Moats

- **Algorithmic Verification Layer:** Independent code-level verification pipeline (exact-quote string matchers, bracket handlers, dangling citation removal) operating downstream of inference.
- **Curated Jurisdiction Legal Packs:** Clean statutory and MPEP regulatory knowledge graphs across US, EPO, UK, and India, eliminating web-search noise and hallucinated case laws.
- **PostgreSQL Row-Level Security (RLS):** 70+ tables guarded by database-layer RLS policies and cryptographic SHA-256 passage provenance. Guaranteed mathematical tenant isolation for competing firms.
- **Microsoft Word Add-in Lock-In:** Deep workflow integration directly inside Microsoft Word (where patent attorneys spend their workdays) creates steep switching costs.
- **Published Empirical Benchmark Leadership:** Open benchmark methodology and failure triage corpora (benchmarks/, evals/) establish SallyIP as the recognized empirical standard of legal AI trust.

5. Market Sizing: A \$65B Global IP Opportunity

TOTAL ADDRESSABLE MARKET (TAM) \$65.8 Billion Global Intellectual Property Services (Filing, Prosecution, Trademark, Litigation).	SERVICEABLE ADDRESSABLE MARKET (SAM) \$4.8 Billion Global LegalTech & Patent Analytics Software (16.4% CAGR to 2030).	SERVICEABLE OBTAINABLE MARKET (SOM) \$920 Million US, UK, and European IP Boutiques & Enterprise Corporate Patent Depts.
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6. Business Model & High-Margin Unit Economics

Tier	Target Customer	Annual Contract Value (ACV)	Key Features Included
Solo Practitioner	Solo patent attorneys & agents	\$1,188 / yr (\$99/mo)	50 scans/mo, claim studio, antecedent checks, DOCX/XML.
IP Boutique & Firm	Mid-sized IP practices (5-30 attys)	\$3,588 / atty / yr (\$299/mo)	Unlimited scans, shared vaults, EoU claim charts, team collaboration.
Enterprise Department	Am Law 100 & Corporate Tech Counsel	\$25,000 - \$150,000+ / yr	Dedicated VPC, BYOK encryption, custom style tuning, SAML SSO, SLA.

7. Competitive Landscape: Why SallyIP Wins

Solution	Core Focus	Critical Limitation	SallyIP Advantage
Harvey AI (\$5B Val)	Horizontal BigLaw AI	High hallucination in patent law; lacks Section 101/112 grammar.	100% Citation Integrity Built specifically for patents.
CoCounsel (TR)	General case research	No limitation-by-limitation claim charts or Office Action traversal.	Full Patent Lifecycle Drafting to USPTO XML export.
PatentPal / ClaimMaster	Rule-based claim checkers	Shallow rules; lacks deep prior art synthesis and conversational interview.	Advanced Generative AI Hybrid search + 7-slot interviewer.
Anaqua / Derwent	Legacy IP docketing	Outdated UI; no generative drafting or verification gates.	10x Faster Cloud Native Immediate practitioner value.

8. Three-Year Financial Forecast

Metric	Year 1 (FY 2027)	Year 2 (FY 2028)	Year 3 (FY 2029)
Ending Annual Recurring Revenue (ARR)	\$1,450,000	\$5,800,000	\$18,200,000
Total Subscribing Attorneys / Seats	420 seats	1,650 seats	5,100 seats
Enterprise Accounts (\$50k+ ACV)	8 enterprise clients	32 enterprise clients	110 enterprise clients
Gross Margin	78%	81%	84%
Net Revenue Retention (NRR)	122%	135%	142%
CAC Payback Period	6.2 months	4.8 months	3.9 months

9. The Ask & Strategic Use of Proceeds

Allocation Area	Budget Share	Key Strategic Deliverables
Engineering & R&D;	40% (\$1.6M)	Scale pgvector to 150M+ patents; fine-tune domain verification SLMs; expand EPC/UK/Asian packs.
Enterprise Sales & GTM	30% (\$1.2M)	Hire 3 enterprise LegalTech Account Execs; attend AIPLA/IPO conferences; scale customer success.
Security & Compliance	20% (\$0.8M)	Attain SOC 2 Type II & ISO 27001 certifications; dedicated enterprise VPC and staging RLS deployment.
Working Capital & IP	10% (\$0.4M)	File core patents covering verification architecture; general corporate & legal reserve.

10. Investment Summary: Why SallyIP Wins

1. **Unforgiving Market Demands Precision:** IP law is the ultimate crucible for legal AI. By solving hallucination where stakes are highest, SallyIP establishes a technical standard general models cannot match.
2. **Defensible Technology Moat:** Code-level verification algorithms, patent-specific statutory gates, and tenant-isolated PostgreSQL RLS protect against horizontal LLM commoditization.
3. **Compelling Unit Economics:** High ACVs (\$18k-\$150k+), high gross margins (>80%), and low churn driven by deep matter knowledge vault and Word add-in integration.
4. **Empirical Truth:** Published benchmarks and transparent failure analysis position SallyIP as the single trusted standard in legal automation.

INVESTOR DUE DILIGENCE & DATA ROOM ACCESS	PRODUCT & ARCHITECTURE REPOSITORY
Web: https://sallyip.com/ Data Room: Available upon NDA execution. Contact: founders@sallyip.com	Technical Doc: about79.md & report29.md Benchmark Corpus: benchmarks/ & evals/ Status: Pilot Phase Active